

Foundations of Artificial Intelligence - 2026.1

Sequential Decisions under Risk - Policy Evaluation

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Decision Making Situations

Decision Situation Matrix

		Action Effects		
Horizon		Certainty	Risk	Uncertainty
	Atomic	✓	✓	
	Sequential	✓	NOW: 25%	

Existence of Optimal Policies

We will suppose here that the MDP is finite (finite S and A).

Under this condition, there must be at least one optimal policy π^* , i.e., a maximizer of the expected utility.

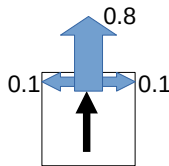
If we want to find $\arg \max_{\pi} \mathbb{E}^{s_0, \pi} [\sum_{t=0}^{\infty} \gamma^t r(S_t)]$, then we first should be able to compute this term for any concrete π .

State Values and Bellman Equation

Example Problem

Consider again the MDP with $\gamma = 1$ and $r(s) = -0.04$ and π given by the arrows:

	1	2	3	4
3	→	→	→	+1
2	↑		↑	-1
1	↑	←	←	←



What is $\mathbb{E}^{s_0, \pi}[\sum_{t=0}^{\infty} \gamma^t r(S_t)]$ for $s_0 = (1, 1)$?

Class Overview

1. State Values and the Bellman Equation
2. Exact Policy Evaluation
3. Iterative Policy Evaluation

State Values

The term $\mathbb{E}^{s_0, \pi}[\sum_{t=0}^{\infty} \gamma^t r(S_t)]$ can be seen as a number or a function of s_0 , π , or both.

If we fix π , we can write it as a function of s as follows

$$v^{\pi}(s) := \mathbb{E}^{s, \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(S_t) \right]$$

The number $v^{\pi}(s) \in \mathbb{R}$ is called the **value of s under policy π** .

It quantifies how good it is to be in state s , on average, if we apply π .

State Values and Bellman Equation

The Bellman Equation

We can write the value of a policy for a state in closed form:

$$v^{\pi}(s) = r(s) + \gamma \sum_{s' \in S} \mathbb{P}(s'|s, \pi(s)) \cdot v^{\pi}(s')$$

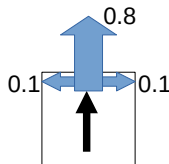
This recursive connection between state values is called the **Bellman equation**.

Observe in particular that $v^{\pi}(s_{\perp}) = 0$ and hence $v^{\pi}(s) = r(s)$ for terminal states.

State Values and Bellman Equation

Example: Using $v^\pi(s) = r(s) + \gamma \sum_{s' \in S} \mathbb{P}(s'|s, \pi(s)) \cdot v^\pi(s')$

	1	2	3	4
3	→	→	→	+1
2	↑		↑	-1
1	↑	←	←	←



We already know that $v^\pi((3,4)) = 1$. Suppose we also know that $v^\pi((2,3)) = 0.66$.

Recalling that $\gamma = 1$ and $r(s) = -0.04$, determine $v^\pi((3,3))$.

Class Overview

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Policy Evaluation

Definition

Policy Evaluation is the task to provide the function $v^\pi(\cdot)$ for a given policy π .

An implicit requirement here is that evaluating $v^\pi(s)$ should be fast for all s .

Once policy evaluation is completed, we can assess the quality of each state, in particular s_0 :

$$\mathbb{E}^{s_0, \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(S_t) \right] = v^\pi(s_0)$$

Policy Evaluation

Policy Evaluation

The Bellman equations yield a system of $|S|$ linear equations and $|S|$ unknowns:

$$v^\pi = r + \gamma \mathbf{P} v^\pi$$

Here, $\mathbf{P} \in \mathbb{R}^{|S| \times |S|}$, and $\mathbf{P}_{ss'} = \mathbb{P}(s'|s, \pi(s))$. For a terminal state s , we have $\mathbf{P}_s = \mathbf{0}$.

If $0 \leq \gamma < 1$, then there is a *unique* solution to this system.

So we can compute all $v^\pi(s)$ simultaneously for a fixed policy π via linear algebra.

Policy Evaluation

Example: State Values for policy π

Solving the system of linear equations gives us the following state values:

	1	2	3	4
3	→	→	→	+1
2	↑		↑	-1
1	↑	←	←	←

	1	2	3	4
3	0.812	0.868	0.918	+1
2	0.762		0.660	-1
1	0.705	0.655	0.611	0.388

Class Overview

1. State Values and the Bellman Equation
2. Exact Policy Evaluation
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Incremental Policy Evaluation

Motivation

There can be several reasons why we cannot solve the set of linear equations in policy evaluation, specifically due to

- memory limitations $\mathcal{O}(|S|^2)$ – unavoidable if we have the matrix $\mathbb{P}(S' | S, \pi(S))$
- time limitations $\mathcal{O}(|S|^3)$
- lack of knowledge about $\mathbb{P}$ or r – only applies under uncertainty

So which other options do we have to compute v^π ?

These problems will not go away as long as we insist on exact values for v^π .

Incremental Policy Evaluation

The Bellman-Update. Idea and Convergence Property.

Instead of computing the exact values v^π , we could opt for **estimates** $\hat{v}^\pi$.

We know that the true state values must satisfy the Bellman equation

$$v^\pi(s) = r(s) + \gamma \sum_{s'} \mathbb{P}(s'|s, \pi(s)) v^\pi(s')$$

Say that at time t we have some idea $\hat{v}_t^\pi$ that does not (yet) satisfy the Bellman equation. How could we try to repair them?

Incremental Policy Evaluation

The Bellman-Update

One idea is to convert the Bellman equation into an **update rule**:

$$\hat{v}_{t+1}^{\pi}(s) \leftarrow r(s) + \gamma \sum_{s'} \mathbb{P}(s'|s, \pi(s)) \hat{v}_t^{\pi}(s')$$

This is a transformative search approach.

Incremental Policy Evaluation

The Bellman-Update is a Contraction

We can think of the Bellman update as an (affine) operator $B : \mathbb{R}^{|S|} \rightarrow \mathbb{R}^{|S|}$.

We write the application of the update rule to state values $x \in \mathbb{R}^{|S|}$ as Bx .

This update operator B in $\mathbb{R}^{|S|}$ is a **contraction**, meaning that, for any values $a, b \in \mathbb{R}^{|S|}$:

$$\|Ba - Bb\|_{\infty} \leq \gamma \|a - b\|_{\infty}$$

Here, $\|x\|_{\infty} := \max_i x_i$ is the **supremum norm**, and $0 < \gamma < 1$.

Incremental Policy Evaluation

The Bellman-Update. Idea and Convergence Property.

If we set $a := \hat{v}_t^\pi$ and $b := v^\pi$ then we get

$$\|\hat{v}_{t+1}^\pi - v^\pi\|_\infty = \|B\hat{v}_t^\pi - Bv^\pi\|_\infty \leq \gamma \|\hat{v}_t^\pi - v^\pi\|_\infty,$$

where $Bv^\pi = v^\pi$ due to the Bellman equation.

The term $\|\hat{v}_{t+1}^\pi - v^\pi\|_\infty$ is the biggest estimation mistake we make for *any* state $s \in S$.

So the update rule reduces the biggest estimation mistake by at least a factor γ .

Incremental Policy Evaluation

The Bellman-Update. Idea and Convergence Property.

Applying the update rule N times from time step t on implies

$$\|\hat{v}_{t+N}^{\pi} - v^{\pi}\|_{\infty} \leq \gamma \|\hat{v}_{t+N-1}^{\pi} - v^{\pi}\|_{\infty} \leq \gamma^2 \|\hat{v}_{t+N-2}^{\pi} - v^{\pi}\|_{\infty} \leq \dots \leq \gamma^N \|\hat{v}_t^{\pi} - v^{\pi}\|_{\infty}$$

So in the limit, we will have that

$$\lim_{N \rightarrow \infty} \|\hat{v}_{t+N}^{\pi} - v^{\pi}\|_{\infty} \leq \lim_{N \rightarrow \infty} \gamma^N \|\hat{v}_t^{\pi} - v^{\pi}\|_{\infty} = (\|\hat{v}_t^{\pi} - v^{\pi}\|_{\infty}) \cdot \underbrace{\lim_{N \rightarrow \infty} \gamma^N}_0 = 0$$

The contraction makes that we converge, from any $\hat{v}_t^{\pi}$, to the true state values v^{π} .

Incremental Policy Evaluation

Incremental Policy Evaluation

We can even show that

$$\|\hat{v}_t^\pi - v^\pi\|_\infty \leq \|\hat{v}_t^\pi - \hat{v}_{t-1}^\pi\|_\infty \cdot \frac{\gamma}{1-\gamma}$$

So if we stop as soon as the (observable) change in the state value estimates is less than $\varepsilon \frac{1-\gamma}{\gamma}$, then we know that the approximation error is below ε .

Incremental Policy Evaluation

Incremental Policy Evaluation

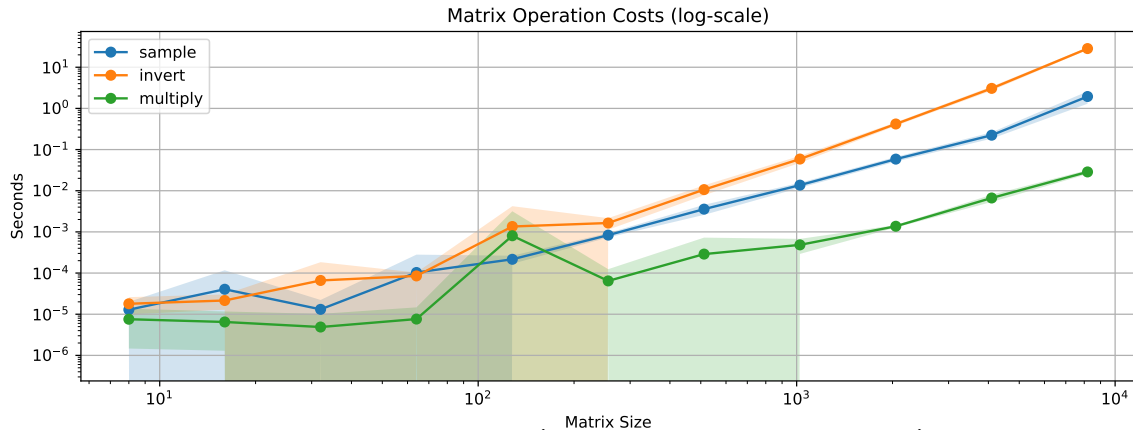
This motivates the following **Incremental Policy Evaluation** algorithm for π :

- 1: Initialize $\hat{v}$ (maybe random, maybe warm-started)
- 2: **repeat**
- 3: $\hat{v}' \leftarrow \hat{v}$
- 4: $\hat{v}(s) \leftarrow r(s) + \gamma \sum_{s'} \mathbb{P}(s'|s, \pi(s)) \hat{v}'(s')$
- 5: **until** $\|\hat{v} - \hat{v}'\|_{\infty} < \varepsilon \frac{1-\gamma}{\gamma}$
- 6: **return** $\hat{v}$

Incremental PE has a runtime of $\mathcal{O}(|S|^2)$ per iteration (matrix-vector multiplication).

Incremental Policy Evaluation

Runtimes



The matrix multiplication is *much* faster (even than sampling a matrix).

We need it multiple times, but it is still often much faster.